



High flow events at Belmont

A report prepared on behalf of Wye Salmon Association (WSA)

Version	2.0
Date	25-Oct-24
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Introduction

This note reports on an analysis of high flow statistics taken from the Belmont gauging station on the river Wye near Hereford. The Belmont gauging station is located at the upstream side of Hereford, thus excluding the Lugg sub catchment. Gauged flow data has been downloaded from the *National River Flow Archive* website and comprises daily average flowrate from 1937 through to 2020. In V2.0, this has been augmented with data for 2021-23 taken from the relatively new [EA Hydrology Data Explorer](#).

The sudden eradication of ranunculus throughout the main stem of the Herefordshire Wye is generally attributed to increased winter flood events. Furthermore, there is often an implication that this is driven by climate change and that things have become significantly worse in the last few years. Examination of the flowrate data paints a rather different story.

High flow events at Belmont

Figure 1a below shows the number of days each year when the river height at Belmont exceeded 4m.

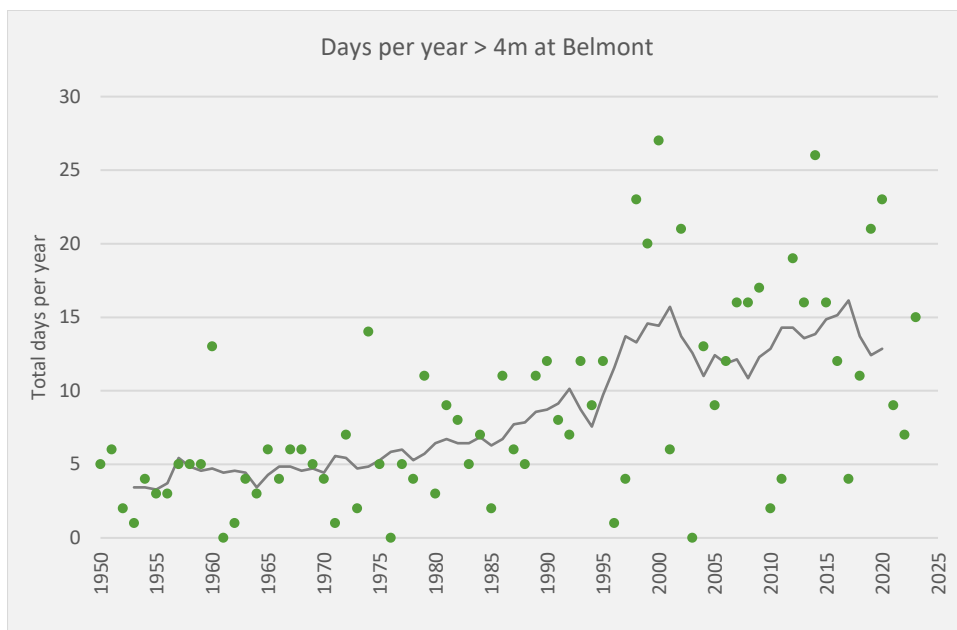


Figure 1a: number of days in each year when daily average flow at Belmont exceeded 230 m³/s, approximating 4m on the height gauge. The trendline is a 7-year rolling average.

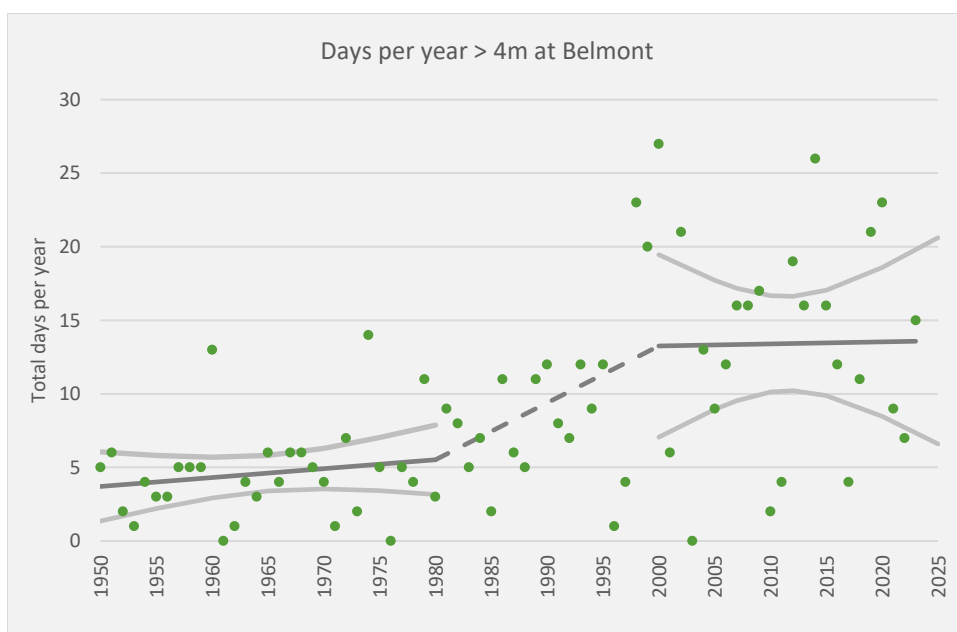


Figure 1b: same data as above. Regression lines and 95% confidence limits are shown separately for two periods: 1950-80 and 2000-23.

The first observation is that there is a huge amount of year-to-year variance – a simple consequence of the variability of the British weather. The trendline, which is a 7-year rolling average, helps expose the underlying trend. Over the period 1950-80, very little changed. Between 1980 and 2000, the number of days over 4m rose rapidly. Since 2000, things have remained stable at the new, higher level.

The two period of relative stability suggests another way of analysing the data. Figure 1b takes the same data and fits a linear regression line to each of the periods 1950-80 and 2000-23. This demonstrates the stability within each of these periods. It also shows that, post 2000, we are seeing approximately three times as many high flow days than was typical between 1950-80; the data variance has also increased by a similar factor. Since 2000, the underlying trend has remained remarkably consistent. Furthermore, there is no hint of increasing variance through the 2000-24 period. Clearly, there is a lot of year-to-year variability. But, as suggested above, this is simply ‘weather’. So, this is already looking very unlike a climate change response. If this were the explanation, surely, we would expect to see things getting substantially worse since 2000. Finally, in the context of ranunculus, there is nothing remarkable about the data around 2019-21 when it disappeared. There were actually more flood days in the period 1998-2002, which it survived perfectly well.

The same story emerges if we examine the *duration* of these high flow events. Figure 2 below shows the duration of the longest period over 4m occurring in each year. As for number of days (figure 1b), the maximum duration data fits neatly onto the same three time periods. During 1950-80, four consecutive days over 4m used to be a once per decade event. Since 2000, it happens often, and six-day durations are not uncommon. As above, the shift occurred between 1980 and 2000, having remained stable since then. Once again, there is nothing remarkable about the period 1019-21, for which maximum flood durations were well below those seen in 2000 and 2014.

Finally, we can examine the frequency of more extreme flooding. Figure 3 overleaf shows the number of days in each year when the river exceeded approximately 5m at Belmont. The picture is similar. 6 years in the 1950s never saw the river exceed 5m at all, now it’s normal – happening every year since 2003. But, as above, the shift occurred between 1980 and 2000 and has not changed significantly over the last 24 years.

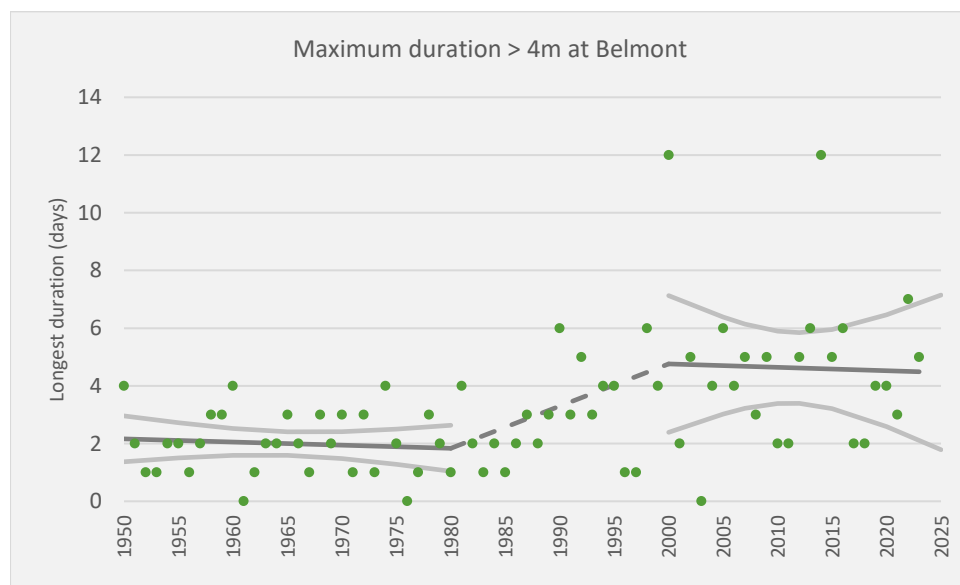


Figure 2: duration of the longest continuous period in each year when daily average flow at Belmont exceeded 230 m³/s, approximating 4m on the height gauge. Regression lines and 95% confidence limits are shown separately for two periods: 1950-80 and 2000-23.

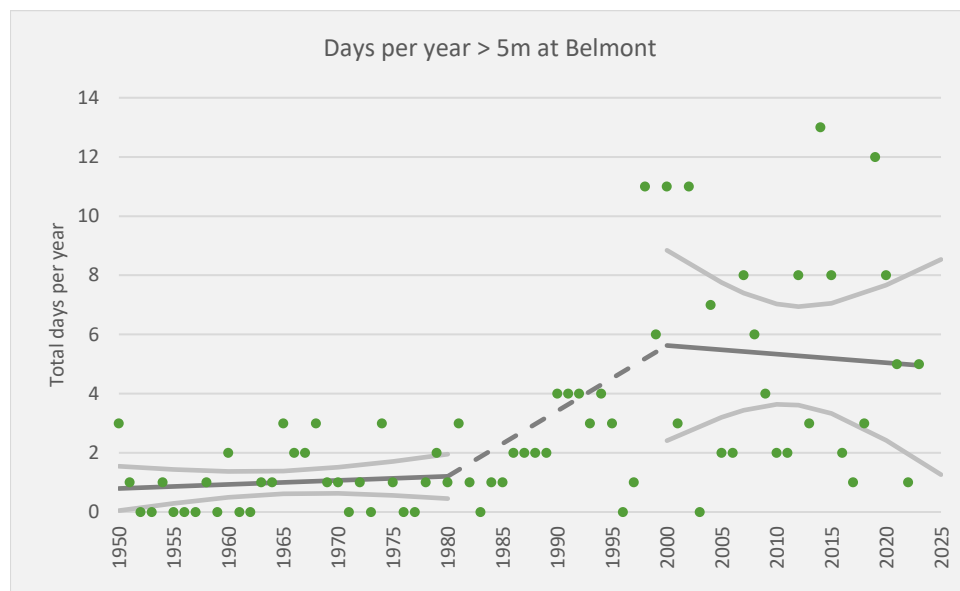


Figure 3: number of days in each year when daily average flow at Belmont exceeded 320 m³/s, approximating 5m on the height gauge. Regression lines and 95% confidence limits are shown separately for two periods: 1950-80 and 2000-23.

Discussion

The pattern within all three datasets of a rapid increase in winter flooding occurring over the period 1980 to 2000, followed by relative stability since then, is not easily attributable to climate change. Some months ago, I discussed this with Jason Doe and Martin Fenn, ecologists at EA. Martin knew exactly what I was referring to and said that the shift was due to 'land use change' during the eighties and nineties. In the largely rural Wye catchment, this phrase essentially equates to changes in rural land management.

This period followed on from the UK joining the EU. As well as the more general pressures to specialise and industrialise farms, the CAP subsidies provided an additional pressure to maximise productive land area. This period saw a lot of removal of woodland, grubbing out hedges, draining boggy land, 'improving' rough grassland, etc. – all of which reduced the ability of the land to retain winter rainfall. The river has essentially been engineered by landowners to be flashier; it comes up faster and comes up higher, in order to drain the land more efficiently.

And ranunculus? I began to ask Martin, "Why did the ranunculus get ripped out by winter floods in 2020 and ...". Martin finished the sentence for me, "... and not in 2000? We don't know either". Clearly something else has been stressing the ranunculus, maybe weakening the root systems. Synergistic effects are not uncommon in complex ecosystems and algal blooms have been suggested as a factor. One might also suggest agricultural pesticides, many of which have increased massively in application volume over the last 20 years. Increased siltation of the gravel caused by soil erosion may also be a factor. But winter floods? Not by themselves.

Before completely dismissing climate change as a driver, it is worth looking at rainfall patterns. Figure 4 shows cumulative winter rainfall for the sub catchment above Belmont.

The regression line suggests that winter rainfall has been increasing at around 2% per decade over the period 1950 to 2020. This is what climate change looks like, a long-term trend. However, the 95% confidence limits (light grey curves) encompass zero gradient so, strictly, we cannot exclude the null hypothesis (i.e. that of no change). What is clear though, is that the underlying rainfall trend is slow moving and small compared to the year-to-year variance (i.e. weather). Importantly, there is no hint of anything anomalous between 1980 and 2000. The rapid change in winter flood patterns in this period was clearly not being driven by rainfall patterns.

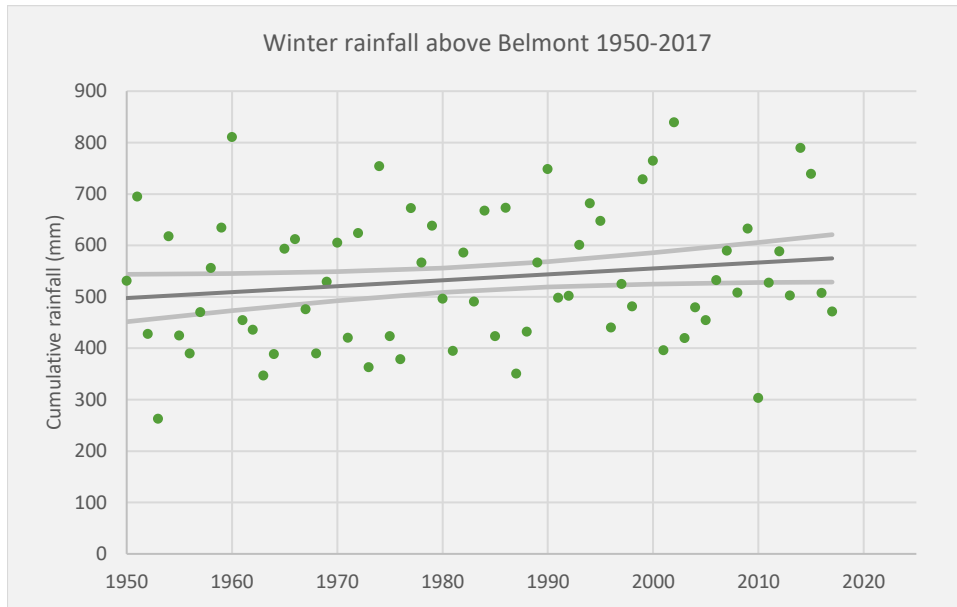


Figure 4: cumulative winter rainfall (Jan, Feb, Nov & Dec) each year. Linear regression line and 95% confidence limits are also shown.

Conclusions

- High flow events on the river Wye above Belmont **increased significantly through the period 1980-2000**, being substantially constant both before and after this period. The river at Belmont now exceeds 4m three times as often as it before 1980. For 5m exceedance, the factor is five times.
- The cause of this rapid increase in winter high flow events is thought to be changes in agricultural land use which have reduced the ability of the land to retain water. As such, this phenomenon should in principle be reversible. **The problem is not the rain – it is how the catchment is currently responding to the rain.**
- The high flow event data (figures 1-3) and the rainfall data (figure 4) together support the view that **climate change is not yet a major driver** of flooding on the Wye (which is not to say that it will not be in the future).
- The relative constancy of winter high flow events since around 2000 throws serious doubt on the suggestion that these events are the sole cause of the loss of ranunculus. Additional factors must be involved.